Exploring Protein Structure Dynamics through Graph-Based Attention and Voronoi Tessellation Saliency Mapping

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Introduction & Summary

Objective: The primary objective of this research is to enhance the understanding of protein-protein interaction (PPI) interfaces by applying Voronoi tessellation and Graph Attention Networks (GATs) to analyze the influence of amino acids and structural properties on PPI stability. This study aims to validate whether these computational methods can effectively identify and quantify crucial structures at protein-protein interfaces, thereby improving predictions of protein stability and interaction dynamics.

Methodology: In this study, we utilized Voronoi tessellation (shown in Fig 3) instead of classical techniques (shown in Fig 6) to transform protein structures into graph representations (shown in Fig 4), where each node corresponds to an amino acid and edges reflect the spatial and interactional dynamics between them. The research employed a robust dataset of 1,567 protein structures derived from rigid-body docking simulations to train our model (shown in Table 1), which integrates Graph Attention Networks (shown in Fig 3) to scrutinize the complex interdependencies within protein complexes.

Additionally, saliency analysis was implemented to identify the most influential amino acids affecting the model's predictions by calculating the gradient of the model's output relative to its inputs, highlighting key contributors to protein stability. We also evaluated Voronoi energy across these structures to understand the correlation between lower energy configurations and increased structural stability, providing insights into the spatial and energetic factors that govern protein interactions. This comprehensive methodology not only enhances the predictability and interpretability of protein stability but also aids in elucidating the fundamental aspects of protein interaction dynamics crucial for advancements in computational biology and protein engineering.

Methods

Our method consists of five primary stages (depicted in Fig. 1) to generate biological insightful findings.

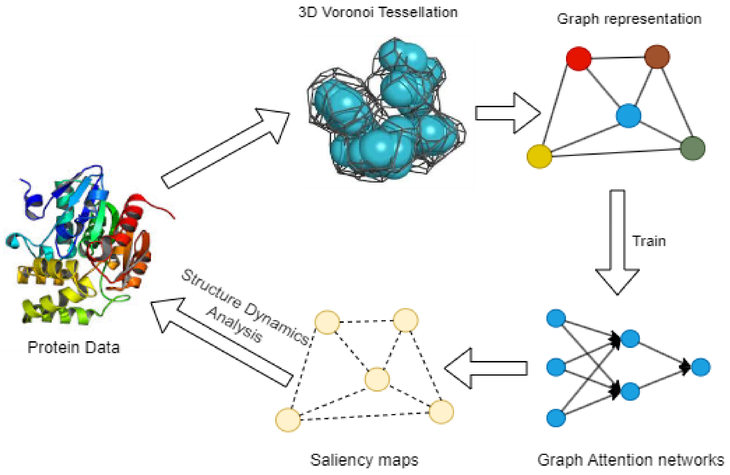


Fig 1: Schematic view of the methodology.

Data set	Targets	Models
Training	1097	14400
Validation	235	2845
Testing	235	2814

Table 1: Dataset training split.

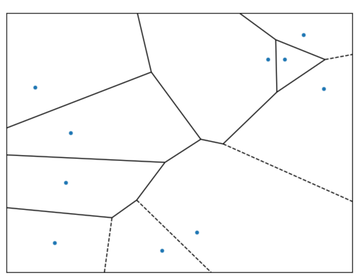


Fig 2: 2D Voronoi representation.

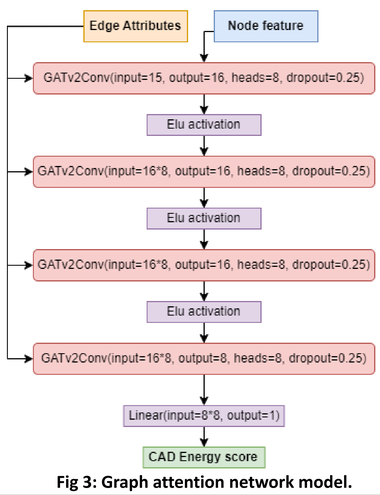


Fig 3: Graph attention network model.

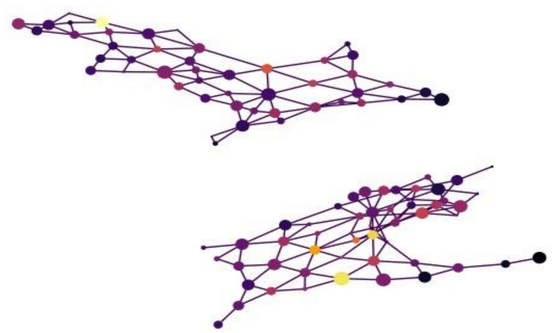
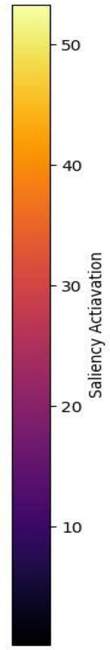


Fig 4 Voronoi tessellation generated protein graph representation.



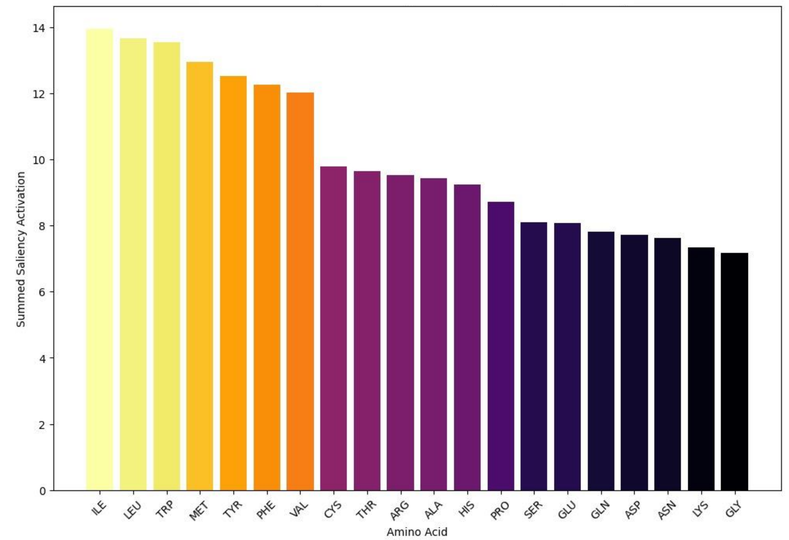


Fig 5: Normalized saliency per amino acid.

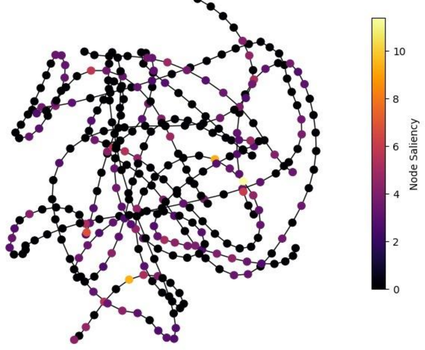


Fig 6: Traditional protein graph representation.

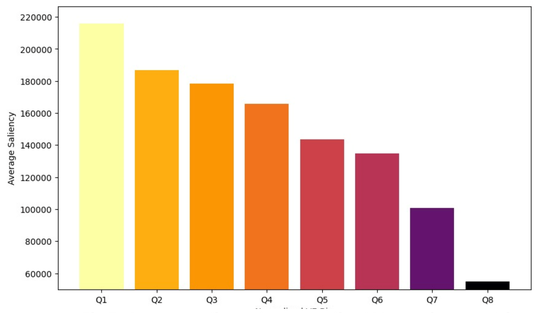


Fig 7: Average saliency by normalized Voronoi energy bin.

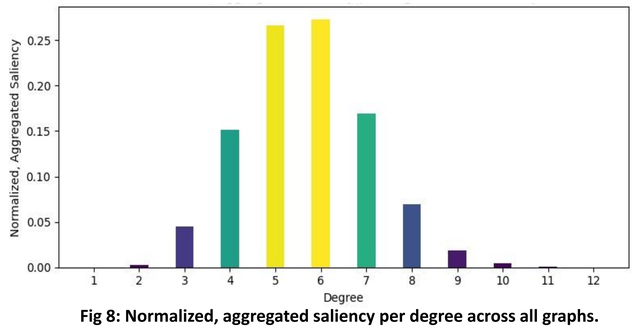


Fig 8: Normalized, aggregated saliency per degree across all graphs.

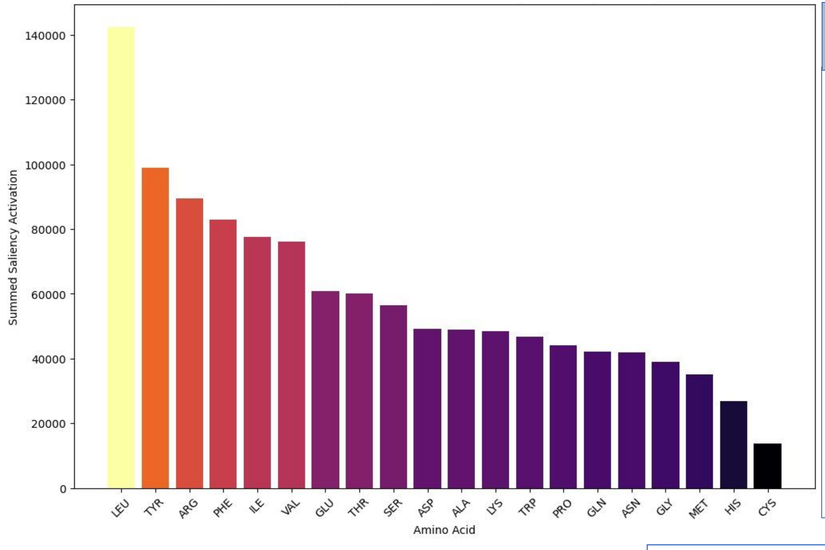


Fig 9: Saliency activation by amino acid.

Biological Insight

Our saliency analysis depicted in Figs. 5 and 9 provides detailed insights into the key factors that contribute to the stability of protein interfaces. Through the application of the graph attention network model, we observed that hydrophobic amino acids, specifically isoleucine and leucine, play a pivotal role in enhancing protein stability. These amino acids are significantly more influential than Glycine, with Isoleucine and Leucine contributing to structural stability at levels 81.48% and 80.77% higher, respectively. This stark contrast underscores the importance of hydrophobic interactions in forming a stable hydrophobic core that is essential for proper protein folding.

Glycine, despite its role in providing structural flexibility, was found to be less impactful in maintaining interface stability. Its smaller side chain results in less extensive van der Waals interactions, which limits its ability to contribute to the stability of protein complexes. This observation aligns with the broader understanding of Glycine's role in protein dynamics, where its flexibility is more crucial in accommodating tight turns or loops rather than stabilizing interfaces.

Moreover, the chart shown in Fig. 7 of Voronoi energy within the protein structures we discovered that lower Voronoi energy regions—indicative of more compact atomic packing—are associated with higher structural stability. This correlation becomes particularly significant in Fig. 8 in which nodes of degrees five and six within the protein interaction graphs. These nodes displayed saliency levels approximately 95.65% and 86.36% higher than those of nodes with lesser degrees of interaction, respectively. Such findings emphasize that nodes with optimal degrees of interaction not only avoid steric hindrance but also enhance stabilization interactions effectively.

The saliency mapping, facilitated by Voronoi tessellation and deep learning techniques, allows for a nuanced mapping of amino acid interdependencies and structural features within protein complexes. This level of detail provides profound insights into the structural dynamics governing protein stability, enhancing the interpretability of computational models. By leveraging these detailed saliency analyses, our research contributes significantly to the targeted design of proteins with improved stability, facilitating advancements in areas such as drug design and protein engineering. Understanding these subtle nuances of protein architecture through saliency analysis is crucial for developing strategies that can manipulate protein behavior and function at the molecular level.

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